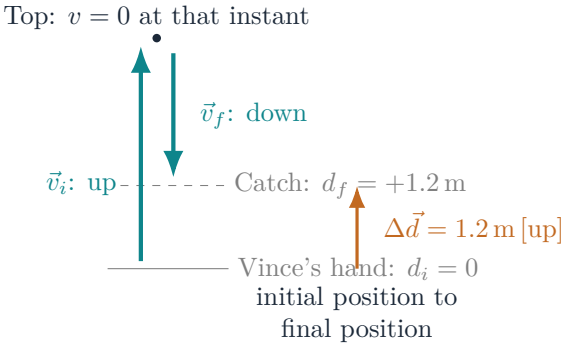
The flying ball: position and displacement

THE FLYING BALL
Vince throws a ball straight up. It rises to its maximum height and then Simeon catches it 1.2 m above Vince's hand on its way back down.
List all the known information about the ball's motion. If the info is not given, leave it blank.

Original motion and the one known change in position



1 Choose an origin and a positive direction

Let Vince's hand be $d_i = 0$ and take up as positive. Simeon catches the ball at $d_f = +1.2\,\text{m}$ relative to that origin. The height above the hand is a position; the ball travels farther than 1.2 m because it first rises and then falls.

2 Separate known directions from unknown magnitudes

Quantity	Known information	Quantity	Known information
d		$\vec{a}$	
$\vec{d}$	$\vec{d}_f = 1.2\,\text{m}\,[\text{up}]$	$\vec{v}_i$	up; size not given
$\Delta\vec{d}$	$1.2\,\text{m}\,[\text{up}]$	$\vec{v}_f$	down; size not given
$\vec{v}_{\text{av}}$	up; size not given	$\Delta\vec{v}$	down; size not given

3 Explain the blank numerical entries

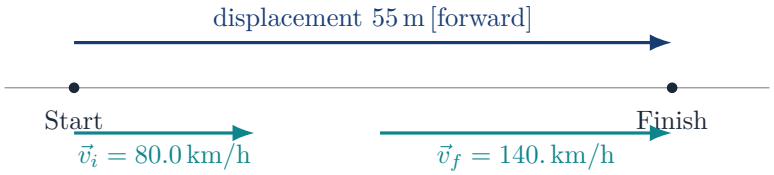
The maximum height and total path length are not given, so d cannot be found. The average velocity points up because net displacement is up. The change in velocity points down because final velocity is down and initial velocity is up. Their sizes need the elapsed time or velocity magnitudes. A numerical acceleration requires an added assumption, such as neglecting air resistance and using $g = 9.8\,\text{m/s}^2$ downward.

Final: At the catch, $\vec{d}_f = \Delta\vec{d} = 1.2\,\text{m}\,[\text{up}]$. The initial velocity points up; the final velocity and its change point down. The remaining numerical entries are undetermined.

Average velocity and change in velocity

THE DIFFERENCE BETWEEN $\vec{v}_{\text{av}}$ AND $\Delta\vec{v}$
A Porsche Carrera accelerates from 80.0 km/h to 140. km/h in 1.8 s and covers 55 m.
Find $\vec{v}_{\text{av}}$ and $\Delta\vec{v}$.

Both results concern the same trip, but measure different changes



elapsed time 1.8 s; choose forward as positive

1 Average velocity uses displacement and elapsed time

Assume that the stated 55 m covered is forward displacement along the road. It is the change in position, not the change in the speedometer reading.

$$\vec{v}_{\text{av}} = \frac{\Delta\vec{d}}{\Delta t}$$

$$\vec{v}_{\text{av}} = \frac{55\,\text{m [forward]}}{1.8\,\text{s}}$$

$$\vec{v}_{\text{av}} = 30.555\ldots\,\text{m/s [forward]}$$

2 Change in velocity subtracts the initial velocity from the final velocity

Both given velocities point forward, so subtract their signed magnitudes.

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i$$

$$\Delta\vec{v} = (140. - 80.0)\,\text{km/h [forward]}$$

$$\Delta\vec{v} = 60.\,\text{km/h [forward]}$$

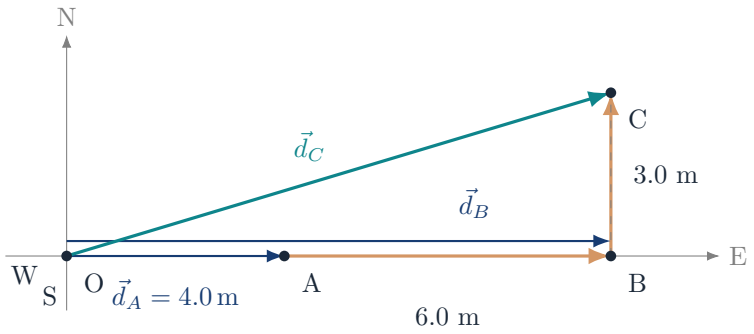
3 Compare the units

Average velocity has units of distance per time. Change in velocity has units of velocity. The car's 60. km/h increase is about 17 m/s, but it is not an average velocity.

Final: $\vec{v}_{\text{av}} = 31\,\text{m/s [forward]}$; $\Delta\vec{v} = 60.\,\text{km/h [forward]}$ (about 17 m/s).

Position of A, B and C from the origin

The original route, with three position vectors drawn from O



In the above diagram, use the origin O as the reference point. A student starts at A, walks to B in 6.0 s then walks to C in another 6.0 s.

POSITION

1. State the position of the student at (Answers at bottom of next page)
   A B C (This one needs to be calculated)

1 Positions all begin at O

A is 4.0 m east of O. B is 4.0 m plus 6.0 m east of O, so its east coordinate is 10.0 m. C is directly north of B: its coordinates measured from O are $(10.0\,\text{m}\,[\text{E}],\ 3.0\,\text{m}\,[\text{N}])$.

2 Find the length of C's position vector

The east and north coordinates form perpendicular legs, so use the Pythagorean theorem.

$$|\vec{d}_C| = \sqrt{(10.0\,\text{m})^2 + (3.0\,\text{m})^2}$$

$$|\vec{d}_C| = 10.440\ldots\ \text{m}$$

3 State its direction from east toward north

$$\theta = \tan^{-1}\left(\frac{3.0}{10.0}\right) = 16.699\ldots^\circ$$

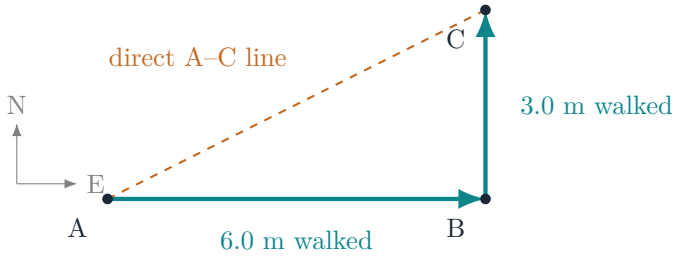
The angle begins at east because the adjacent leg of the triangle points east.

Final: $\vec{d}_A = 4.0\,\text{m}\,[\text{E}]$; $\vec{d}_B = 10.0\,\text{m}\,[\text{E}]$; $\vec{d}_C = 10.4\,\text{m}\,[\text{E}17^\circ N]$.

Distance walked along the path

DISTANCE
2. What is the distance walked as the student walks from
A to B B to C A to C

Distance follows the two sections of the student's actual walk



1 Read the length of each section of the path

From A to B, the student walks 6.0 m. From B to C, the student walks another 3.0 m. Distance is a scalar, so write lengths in metres without compass directions.

2 Add the two sections for the entire trip

$$d_{A\to C} = d_{A\to B} + d_{B\to C}$$
$$d_{A\to C} = 6.0\,\text{m} + 3.0\,\text{m}$$
$$d_{A\to C} = 9.0\,\text{m}$$

3 Check which line you measured

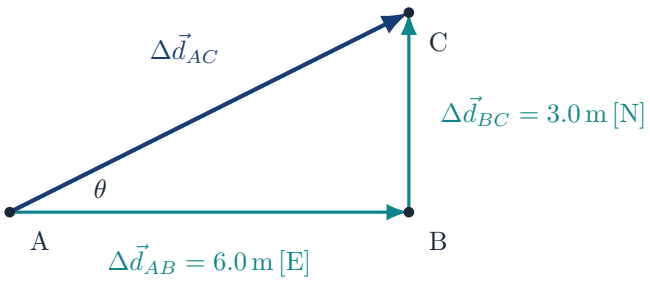
The dashed diagonal connects A and C directly. The student did not walk along it, so its length is not the distance travelled. That straight-line length will be the magnitude of displacement in question 3.

Final: A to B: 6.0 m; B to C: 3.0 m; A to C: 9.0 m.

Add the displacements from A to C

DISPLACEMENT
3. What is the displacement as the student walks from
   A to B B to C A to C

Tip-to-tail construction: the diagonal is the total displacement



1 State the displacement of each leg

From A to B, the change in position is 6.0 m east. From B to C, it is 3.0 m north. These are vectors: direction is part of each answer.

2 Connect the first tail to the final tip

$$\Delta\vec{d}_{AC} = \Delta\vec{d}_{AB} + \Delta\vec{d}_{BC}$$

The two leg vectors meet at B; their sum runs directly from A to C. This is a right triangle with an east leg of 6.0 m and a north leg of 3.0 m.

$$|\Delta\vec{d}_{AC}| = \sqrt{(6.0\,\text{m})^2 + (3.0\,\text{m})^2}$$

$$|\Delta\vec{d}_{AC}| = 6.708\ldots\,\text{m}$$

3 Measure the angle from east toward north

$$\theta = \tan^{-1}\left(\frac{3.0}{6.0}\right) = 26.565\ldots^\circ$$

The north leg is opposite this angle and the east leg is adjacent. Both directions are positive, so the resultant points northeast.

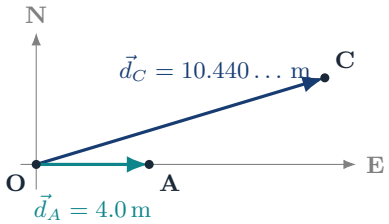
Final: $\Delta\vec{d}_{AB} = 6.0\,\text{m}\,[\text{E}]$; $\Delta\vec{d}_{BC} = 3.0\,\text{m}\,[\text{N}]$;
$\Delta\vec{d}_{AC} = 6.7\,\text{m}\,[\text{E}27^\circ N]$.

Reverse the position vector being subtracted

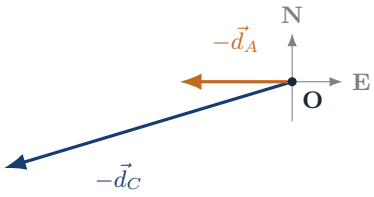
4. Try it. Take your final position at C (found above) and your initial position at A and do a vector subtraction. You should get the same total displacement you got above from A to C.

The two position vectors from O and their opposites

Original position vectors



Opposite vectors from a common tail

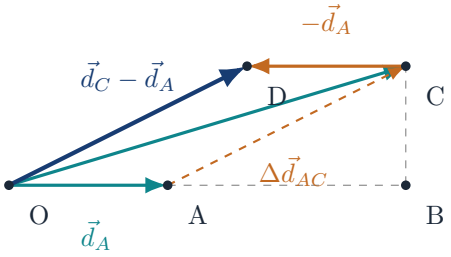


1 Keep the vectors' lengths when reversing them

$\vec{d}_A$ points east, so $-\vec{d}_A$ points west with the same 4.0 m magnitude. $-\vec{d}_C$ points opposite $\vec{d}_C$, toward the southwest. Only $-\vec{d}_A$ is needed because the expression subtracts $\vec{d}_A$.

$$\Delta\vec{d}_{AC} = \vec{d}_C - \vec{d}_A = \vec{d}_C + (-\vec{d}_A)$$

For subtraction, move $-\vec{d}_A$ to the tip of $\vec{d}_C$ without turning it

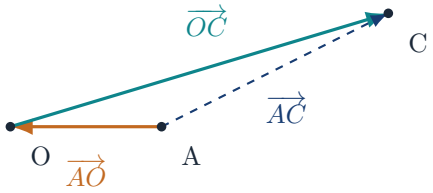


2 Read the tip-to-tail result

Start with $\vec{d}_C$ from O to C. Place $-\vec{d}_A$ from C west to the construction point D. The solid resultant O to D and dashed displacement A to C have the same length and direction. D is not a stop on the walk.

Algebraic method for subtracting positions

Question 4 continued: rewrite the subtraction as a tip-to-tail path



1 Name the final and initial position vectors

The final position $\vec{d}_C$ is $\overrightarrow{OC}$; the initial position $\vec{d}_A$ is $\overrightarrow{OA}$. Subtract the initial position from the final one.

$$\Delta\vec{d}_{AC} = \vec{d}_C - \vec{d}_A$$

$$\Delta\vec{d}_{AC} = \overrightarrow{OC} - \overrightarrow{OA}$$

2 Reverse the arrow that is being subtracted

The negative of $\overrightarrow{OA}$ is $\overrightarrow{AO}$: it has the same length and points west instead of east.

$$\Delta\vec{d}_{AC} = \overrightarrow{OC} + \overrightarrow{AO}$$

3 Reorder the vectors to make a continuous path

Vector addition is commutative. Put $\overrightarrow{AO}$ first, then place $\overrightarrow{OC}$ at its tip. This traces A to O to C. The resultant goes directly from A to C.

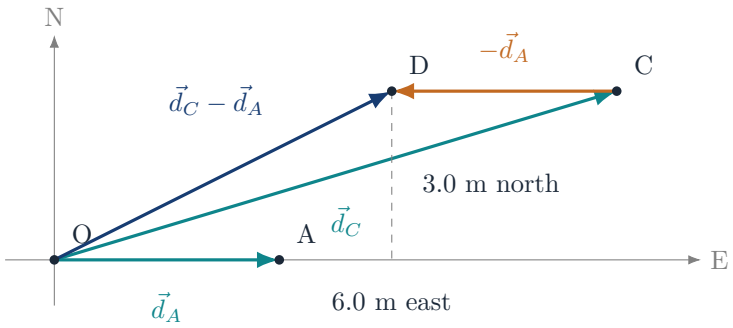
$$\Delta\vec{d}_{AC} = \overrightarrow{AO} + \overrightarrow{OC}$$

$$\Delta\vec{d}_{AC} = \overrightarrow{AC}$$

Final: Algebraic method: $\vec{d}_C - \vec{d}_A = \overrightarrow{AC}$, the displacement from A to C. The coordinate method on the next page gives its length and direction.

Coordinate method for subtracting positions

Question 4 continued: subtract east and north coordinates separately



1 Write both original position vectors on the same axes

Choose east as $+x$ and north as $+y$. Position A is 4.0 m east of O, and position C is 10.0 m east and 3.0 m north of O.

$$\vec{d}_A = (4.0\,\text{m}, 0)$$

$$\vec{d}_C = (10.0\,\text{m}, 3.0\,\text{m})$$

2 Subtract component by component

This is the coordinate form of adding $-\vec{d}_A = (-4.0\,\text{m}, 0)$. Its westward component cancels 4.0 m of C's 10.0 m eastward position.

$$\Delta\vec{d}_{AC} = \vec{d}_C - \vec{d}_A$$

$$\Delta\vec{d}_{AC} = (10.0\,\text{m}, 3.0\,\text{m}) - (4.0\,\text{m}, 0)$$

$$\Delta\vec{d}_{AC} = (6.0\,\text{m}, 3.0\,\text{m})$$

3 Find the length and bearing from the signed components

Both components are positive. They make a right triangle with an east leg of 6.0 m and a north leg of 3.0 m.

$$|\Delta\vec{d}_{AC}| = \sqrt{(6.0\,\text{m})^2 + (3.0\,\text{m})^2}$$

$$|\Delta\vec{d}_{AC}| = 6.708203\ldots\ \text{m}$$

$$\theta = \tan^{-1}\left(\frac{3.0}{6.0}\right) = 26.565051\ldots^\circ$$

Final: Coordinate method: $\Delta\vec{d}_{AC} = 6.7\,\text{m}\,[\text{E}27^\circ N]$. Both methods give the result from question 3.

Average speed on each leg and the whole trip

AVERAGE SPEED
5. Definition: The rate at which distance is covered. Write the equation for this:

1 Translate "distance per time" into an equation

The numerator is the length of the path walked. The denominator is the time spent walking that path. Neither is a vector.

$$v_{\text{av,speed}} = \frac{d_{\text{total}}}{\Delta t}$$

6. Find the average speed of the student when walking from (Answers at bottom)
   A to B B to C A to C

Distance and time must cover the same interval



Full trip A to C: 9.0 m in 12.0 s

2 Calculate speed from A to B

$$v_{\text{av,speed},AB} = \frac{6.0\,\text{m}}{6.0\,\text{s}} = 1.0\,\text{m/s}$$

3 Calculate speed from B to C

$$v_{\text{av,speed},BC} = \frac{3.0\,\text{m}}{6.0\,\text{s}} = 0.50\,\text{m/s}$$

4 Use the entire path and entire time for A to C

The path length is $6.0 + 3.0 = 9.0\,\text{m}$. The elapsed time is $6.0 + 6.0 = 12.0\,\text{s}$. Dividing the straight-line displacement by time would answer an average *velocity* question instead.

$$v_{\text{av,speed},AC} = \frac{9.0\,\text{m}}{12.0\,\text{s}} = 0.75\,\text{m/s}$$

Final: A to B: 1.0 m/s; B to C: 0.50 m/s; A to C: 0.75 m/s.

Average velocity on the east and north legs

AVERAGE VELOCITY
7. Definition: The rate of change of position. Write the equation for this:

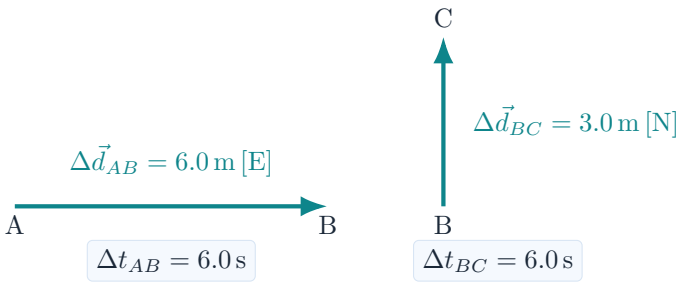
1 **Use displacement, a vector, instead of distance**

$$\vec{v}_{\text{av}} = \frac{\Delta \vec{d}}{\Delta t}$$

Here $\Delta \vec{d} = \vec{d}_f - \vec{d}_i$. The time is scalar, so dividing by a positive time preserves the displacement's direction.

8. Find the average velocity of the student when walking from
   A to B B to C A to C

The first two intervals are straight, so each velocity points along its leg



2 **Calculate average velocity from A to B**

$$\vec{v}_{\text{av},AB} = \frac{6.0\,\text{m}\,[\text{E}]}{6.0\,\text{s}}$$
$$\vec{v}_{\text{av},AB} = 1.0\,\text{m/s}\,[\text{E}]$$

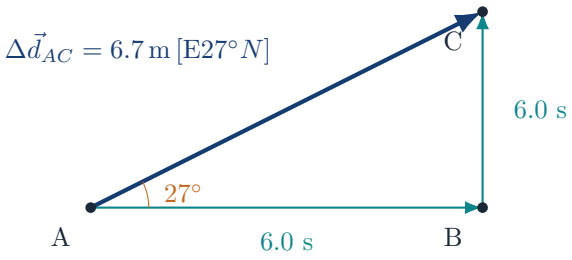
3 **Calculate average velocity from B to C**

$$\vec{v}_{\text{av},BC} = \frac{3.0\,\text{m}\,[\text{N}]}{6.0\,\text{s}}$$
$$\vec{v}_{\text{av},BC} = 0.50\,\text{m/s}\,[\text{N}]$$

Final: Question 7: $\vec{v}_{\text{av}} = \Delta \vec{d}/\Delta t$. Question 8 so far: A to B is $1.0\,\text{m/s}\,[\text{E}]$; B to C is $0.50\,\text{m/s}\,[\text{N}]$. The full trip is continued on the next page.

Average velocity for the full trip and the meaning of rate

Question 8 continued: A to C uses the direct displacement and the full elapsed time



4 Divide A-to-C displacement by A-to-C time

Question 3 found a direct displacement of $6.708\ldots$ m at $26.565\ldots°$ north of east. Both 6.0 s legs count toward the time from A to C: $\Delta t_{AC} = 12.0\,\text{s}$. Dividing by a positive time does not turn the direction.

$$\vec{v}_{\text{av},AC} = \frac{6.708\ldots\,\text{m}\,[\text{E}26.565\ldots°N]}{12.0\,\text{s}}$$

$$\vec{v}_{\text{av},AC} = 0.5590\ldots\,\text{m/s}\,[\text{E}26.565\ldots°N]$$

5 Compare full-trip speed and velocity

The route is 9.0 m long, while the direct displacement is 6.7 m. Both calculations use the same 12.0 s; their numerators differ. This gives average speed 0.75 m/s and average velocity 0.56 m/s [E27°N].

Final: Question 8, A to C: $\vec{v}_{\text{av},AC} = 0.56\,\text{m/s}\,[\text{E}27°N]$.

9. What do you think it means when a definition uses the word “rate”?

1 Look for a quantity divided by elapsed time

The numerator names what accumulates or changes. The denominator tells how long it takes. “Per second” is one way to read a rate: a speed of 0.75 m/s means an average of 0.75 m of path distance for every second of the full trip.

Final: A rate measures how much a quantity changes or accumulates per unit time. Speed is distance per time; velocity is displacement per time.